

Backend Learning — Tools & Technologies

Every concept mapped to the trending framework or tool you should actually learn

Color Legend

Industry standard

Trending / modern pick

Popular alternative

Framework-specific

Cloud / managed

Self-hosted

Utility / CLI

Phase 1 — Foundation Gaps		Days 1–8		
Concept	Tools & Frameworks			
HTTP client	Axios	Fetch API	Got	
API framework	Express.js	FastAPI	NestJS	Hono
Auth / JWT	Auth0	Clerk	Passport.js	Keycloak
OAuth provider	NextAuth	Supertokens	Ory	
Async runtime	Node.js	Bun	Deno	
API testing	Postman	Bruno	HTTPie	
Docs / OpenAPI	Swagger UI	Scalar	Redoc	

★ Trending: Hono (ultra-fast), Bun (runtime), Clerk (auth), Bruno (API testing)

Phase 2 — Database Mastery		Days 9–16		
Concept	Tools & Frameworks			
SQL database	PostgreSQL	MySQL	SQLite	
ORM	Prisma	Drizzle	TypeORM	SQLAlchemy
Query builder	Knex.js	kysely		
Connection pool	PgBouncer	pgpool-II		
Migrations	Prisma Migrate	Flyway	Liquibase	Alembic
NoSQL	MongoDB	DynamoDB	Cassandra	
Cache / KV	Redis	Dragonfly	Memcached	
DB as a service	Neon	PlanetScale	Supabase	
Query analysis	EXPLAIN ANALYZE	pgBadger	pganalyze	

★ Trending: Prisma & Drizzle (type-safe ORMs), Neon (serverless Postgres), Dragonfly (Redis drop-in)

Phase 3 — System Design					Days 17–24
Concept	Tools & Frameworks				
Caching layer	Redis	Cloudflare Cache	Varnish		
Message queue	RabbitMQ	Apache Kafka	SQS	Redpanda	
Background jobs	BullMQ	Celery	Temporal	Inngest	
Rate limiting	Upstash Ratelimit	Redis + Lua	Kong		
API gateway	Kong	Traefik	AWS API GW	NGINX	
Service mesh	Istio	Linkerd	Consul		
CDN	Cloudflare	AWS CloudFront	Fastly		
Search	Elasticsearch	Typesense	Meilisearch		
★ Trending: BullMQ (Node jobs), Temporal (durable workflows), Redpanda (Kafka-compatible), Inngest (event-driven)					

Phase 4 — Reliability & Observability					Days 25–31
Concept	Tools & Frameworks				
Load testing	k6	Artillery	Locust	JMeter	
Profiling	clinic.js	0x	py-spy	Chrome DevTools	
Metrics	Prometheus	Datadog	StatsD		
Tracing	OpenTelemetry	Jaeger	Tempo	Zipkin	
Logging	Pino	Winston	Loki	ELK Stack	
Dashboards	Grafana	Datadog	New Relic		
Circuit breaker	Opossum	Resilience4j	Polly		
Error tracking	Sentry	Highlight.io			

★ Trending: OpenTelemetry (standard for traces), Pino (fastest Node logger), k6 (code-first load tests), Loki (log storage)

Phase 5 — Security & Deployment		Days 32–37		
Concept	Tools & Frameworks			
Secrets mgmt	HashiCorp Vault	Doppler	AWS Secrets Mgr	
Input validation	Zod	Pydantic	Joi	class-validator
Container	Docker		Podman	
Orchestration	Kubernetes	ECS	Nomad	
CI/CD	GitHub Actions	GitLab CI	CircleCI	
Feature flags	LaunchDarkly	Unleash	PostHog	
Security scan	Snyk	Trivy	Dependabot	
IaC	Terraform	Pulumi	CDK	

★ Trending: Zod (validation everywhere), Doppler (secrets), Trivy (container scanning), Pulumi (IaC in code)

Phase 6 — Capstone Practice					Days 38–40	
Concept	Tools & Frameworks					
Deploy platform	Railway		Fly.io		Render	AWS / GCP
Event streaming	Kafka		Redpanda		EventBridge	
Real-time	Socket.io		Ably		Pusher	
System design	Excalidraw		Eraser.io		Miro	
Documentation	Notion		Obsidian		Confluence	

★ Trending: Railway / Fly.io (easiest deploys), Eraser.io (system design diagrams), Redpanda

Top Trending Picks — Learn These First

Category	Trending Pick	Why
ORM (Node.js)	Drizzle ORM	Type-safe, lightweight, SQL-like syntax, very fast adoption
Background jobs	BullMQ	Redis-backed, battle-tested, great DX for Node.js
Validation	Zod	TypeScript-first, compose schemas, used everywhere
Logging	Pino	10x faster than Winston, structured JSON out of the box
Load testing	k6	Code-first, CI-friendly, Grafana integration
Tracing	OpenTelemetry	Vendor-neutral standard — works with any backend
Auth	Clerk	Best DX, handles sessions, JWTs, OAuth in one SDK
Postgres SaaS	Neon	Serverless Postgres with branching — great for dev/prod parity
Queues	Temporal	Durable workflows, handles retries/state automatically
API framework	Hono	Fastest Node/Bun/Deno framework, edge-native